CM0859 – MT5009 Type Theory
Natural Deduction

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Preliminaries

Textbook

"Lecture Notes on Natural Deduction" (Pfenning 2023).

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Conventions

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Preliminaries

Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	(inclusive) disjunction
$\supset$	(if __, then __)	implication
$\bot$	(falsity)	bottom, falsum
$\top$	(truth)	top
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier

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Logical constants

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$\top$	(truth)	top
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier

Definition

We define negation by $\neg A :\equiv A \supset \perp$.

Formalisation of Logical Systems

In relation to the notions of propositions and proofs the textbook says:

> *"There is no universal agreement about the proper foundations for these notions." (p. L2.1)*

Formalisation of Logical Systems

$$\text{Logical System} \begin{cases} \text{Axiomatic proof theory (Hilbert)} \\ \text{Structural proof theory (Gentzen)} \begin{cases} \text{Natural deduction} \\ \text{Sequent calculus} \end{cases} \end{cases}$$

Formalisation of Logical Systems[†]

$$\text{Logical System}\begin{cases}\text{Axiomatic proof theory (Hilbert)}\\[2ex] \text{Structural proof theory (Gentzen)}\begin{cases}\text{Natural deduction}\\[1ex] \text{Sequent calculus}\end{cases}\end{cases}$$

[†] Complementary material: “Basic Proof Theory” (Troelstra and Schwichtenberg [1996] 2000) and “Structural Proof Theory” (Negri and von Plato [2001] 2008).

Judgements and Propositions

"The cornerstone of Martin-Löf's foundation of logic is a clear separation of the notions of judgment and proposition." (p. L2.1)

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"He ⟦Martin-Löf (1996) and Martin-Löf (1987)⟧ reasons that to judge is to know and that an evident judgment is an object of knowledge. A proof is what makes a judgment evident. In logic, we make particular judgments such as 'A is a proposition' or 'A is true', presupposing in the latter case that A is already known to be a proposition. To know that 'A is a proposition' means to know what counts as a verification of A, whereas to know that 'A is true' means to know how to verify A." (Pfenning and Davies 2001, p. 512)

Judgements and Propositions

Examples (judgements)

- "A is a proposition".
- "The proposition A is true".
- "The proposition A is false".
- "The proposition A is true at time t" (temporal logic).
- "The proposition A is necessarily true" (modal logic).
- "σ is a type" (programming languages).
- "M is a combinator/program" (programming languages).
- "The combinator/program M has type σ" (programming languages).

Judgements and Propositions

From the sharp separation of the notions of judgement and proposition by Martin-Löf, we know that inference rules operate on judgements and logical constants operate on propositions.

Natural Deduction

Introduction

- Propositions: $A, B, C, \ldots$
- Judgements:
  (i) $A\,\mathrm{prop}$ (A is a proposition)
  (ii) $A\,\mathrm{true}$ (the proposition A is true)
- Form of the inference rules:

$$\frac{J_1 \quad \ldots \quad J_n}{J}\ \text{rule name}$$

  where J (conclusion) and $J_1, \ldots, J_n$ (premises) are all judgements.
- Inference rules: Formation, introduction and elimination rules

  > *"If proofs are programs then we need to explain how proofs are to be executed, and which results may be returned by a computation." (p. L4.7)*

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Natural Deduction for Intuitionistic Logic

We shall define inference rules for conjunction, implication, disjunction, truth, falsehood and negation.

Natural Deduction for Intuitionistic Logic

Inference rules for conjunction

Natural Deduction for Intuitionistic Logic

Inference rules for conjunction

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \wedge B\,\mathrm{prop}}$$

Natural Deduction for Intuitionistic Logic

Inference rules for conjunction

- Formation rule

$$\frac{A\,\mathsf{prop} \qquad B\,\mathsf{prop}}{A \wedge B\,\mathsf{prop}}$$

- Introduction rule

$$\frac{A\,\mathsf{true} \qquad B\,\mathsf{true}}{A \wedge B\,\mathsf{true}}\,\wedge\mathrm{I}$$

Natural Deduction for Intuitionistic Logic

Inference rules for conjunction

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \wedge B\,\mathrm{prop}}$$

- Introduction rule

$$\frac{A\,\mathrm{true} \qquad B\,\mathrm{true}}{A \wedge B\,\mathrm{true}}\;\wedge\mathrm{I}$$

- Elimination rules

$$\frac{A \wedge B\,\mathrm{true}}{A\,\mathrm{true}}\;\wedge\mathrm{E}_1 \qquad\qquad \frac{A \wedge B\,\mathrm{true}}{B\,\mathrm{true}}\;\wedge\mathrm{E}_2$$

Natural Deduction for Intuitionistic Logic

Inference rules for implication

Natural Deduction for Intuitionistic Logic

Inference rules for implication

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \supset B\,\mathrm{prop}}$$

Natural Deduction for Intuitionistic Logic

Inference rules for implication

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \supset B\,\mathrm{prop}}$$

- Introduction rule

$$\frac{\dfrac{}{A\,\mathrm{true}}\,u \atop {\vdots \atop B\,\mathrm{true}}}{A \supset B\,\mathrm{true}}\,\supset\mathrm{I}^u$$

In the application of the $\supset\mathrm{I}$ rule, we may discharge zero, one, or more occurrences of the assumption.)

Natural Deduction for Intuitionistic Logic

Inference rules for implication

- Formation rule

$$\frac{A\ \mathrm{prop} \qquad B\ \mathrm{prop}}{A \supset B\ \mathrm{prop}}$$

- Introduction rule

$$\begin{array}{c} \overline{A\ \mathrm{true}}^{\,u} \\ \vdots \\ \dfrac{B\ \mathrm{true}}{A \supset B\ \mathrm{true}}\ {\supset}\mathrm{I}^u \end{array}$$

  In the application of the $\supset\mathrm{I}$ rule, we may discharge zero, one, or more occurrences of the assumption.)

- Elimination rule

$$\frac{A \supset B\ \mathrm{true} \qquad A\ \mathrm{true}}{B\ \mathrm{true}}\ {\supset}\mathrm{E}$$

Natural Deduction for Intuitionistic Logic

Example

A derivation where the assumption and the conclusion are the same and the assumption is discharged once. A proof that $A \supset A\,\mathrm{true}$.

$$\dfrac{\dfrac{}{A\,\mathrm{true}}\,u}{A \supset A\,\mathrm{true}}\,{\supset}\mathrm{I}^{u}$$

Natural Deduction for Intuitionistic Logic

Example

A derivation using the same assumption twice. A proof that $(A \wedge B) \supset (B \wedge A)\,\mathsf{true}$.

$$
\dfrac{\dfrac{\dfrac{}{A \wedge B\ \mathrm{true}}\,u}{B\ \mathrm{true}}\,\wedge E_2 \qquad \dfrac{\dfrac{}{A \wedge B\ \mathrm{true}}\,u}{A\ \mathrm{true}}\,\wedge E_1}{\dfrac{B \wedge A\ \mathrm{true}}{(A \wedge B) \supset (B \wedge A)\ \mathrm{true}}\,\supset I^u}\,\wedge I
$$

Natural Deduction for Intuitionistic Logic

Example

A derivation where there is a vacuous discharge when using the inference rule $\supset\mathrm{I}$. A proof that $A \supset (B \supset A)\,\mathrm{true}$.

$$
\dfrac{\dfrac{\dfrac{}{A\,\mathrm{true}}\,u}{B \supset A\,\mathrm{true}}\,\supset\mathrm{I}\ \text{(vacuous discharge of } A\text{)}}{A \supset (B \supset A)\,\mathrm{true}}\,\supset\mathrm{I}^u
$$

Natural Deduction for Intuitionistic Logic

Notation

Let Γ be a set of judgements and let $A\,\mathrm{true}$ be a judgement. The relation $\Gamma \vdash A\,\mathrm{true}$ means that there is a derivation with conclusion $A\,\mathrm{true}$ from the set of hypotheses Γ.

Natural Deduction for Intuitionistic Logic

Example

A derivation using the same hypothesis twice. A proof that $A \wedge B \,\mathrm{true} \vdash B \wedge A \,\mathrm{true}$.

$$\dfrac{\dfrac{A \wedge B \,\mathrm{true}}{B \,\mathrm{true}}\wedge\mathrm{E}_2 \qquad \dfrac{A \wedge B \,\mathrm{true}}{A \,\mathrm{true}}\wedge\mathrm{E}_1}{B \wedge A \,\mathrm{true}}\wedge\mathrm{I}$$

Inference rules for disjunction

Natural Deduction for Intuitionistic Logic

Inference rules for disjunction

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \vee B\,\mathrm{prop}}$$

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Inference rules for disjunction

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \vee B\,\mathrm{prop}}$$

- Introduction rules

$$\frac{A\,\mathrm{true}}{A \vee B\,\mathrm{true}}\,\vee\mathrm{I}_1 \qquad\qquad \frac{B\,\mathrm{true}}{A \vee B\,\mathrm{true}}\,\vee\mathrm{I}_2$$

Natural Deduction for Intuitionistic Logic

Inference rules for disjunction

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \vee B\,\mathrm{prop}}$$

- Introduction rules

$$\frac{A\,\mathrm{true}}{A \vee B\,\mathrm{true}}\,\vee\mathrm{I}_1 \qquad\qquad \frac{B\,\mathrm{true}}{A \vee B\,\mathrm{true}}\,\vee\mathrm{I}_2$$

- Elimination rule

$$\frac{A \vee B\,\mathrm{true} \qquad \begin{array}{c}\overline{A\,\mathrm{true}}^{\,u} \\ \vdots \\ C\,\mathrm{true}\end{array} \qquad \begin{array}{c}\overline{B\,\mathrm{true}}^{\,w} \\ \vdots \\ C\,\mathrm{true}\end{array}}{C\,\mathrm{true}}\,\vee\mathrm{E}^{u,w}$$

Natural Deduction for Intuitionistic Logic

Inference rules for truth

Natural Deduction for Intuitionistic Logic

Inference rules for truth

- Formation rule

$$\frac{}{\top \; \mathrm{prop}}$$

Natural Deduction for Intuitionistic Logic

Inference rules for truth

- Formation rule

$$\frac{}{\top\ \mathrm{prop}}$$

- Introduction rule

$$\frac{}{\top\ \mathrm{true}}\ \top I$$

Natural Deduction for Intuitionistic Logic

Inference rules for truth

- Formation rule

$$\frac{}{\top \text{ prop}}$$

- Introduction rule

$$\frac{}{\top \text{ true}}\ \top I$$

- Elimination rule

  "We have no information if we know $\top$ true*, so there is no elimination rule." (p. L2.2)*

Natural Deduction for Intuitionistic Logic

Inference rules for falsehood

Natural Deduction for Intuitionistic Logic

Inference rules for falsehood

- Formation rule

$$\frac{}{\bot \ \mathrm{prop}}$$

Natural Deduction for Intuitionistic Logic

Inference rules for falsehood

- Formation rule

$$\frac{}{\bot \ \mathrm{prop}}$$

- Introduction rule

  "$\bot$ is a proposition that should have no proof! Therefore there are no introduction rules." (p. L2.10)

Natural Deduction for Intuitionistic Logic

Inference rules for falsehood

- Formation rule

$$\frac{}{\bot \; \mathrm{prop}}$$

- Introduction rule

  "$\bot$ is a proposition that should have no proof! Therefore there are no introduction rules." (p. L2.10)

- Elimination rule

$$\frac{\bot \; \mathrm{true}}{C \; \mathrm{true}} \; \bot\mathrm{E}$$

Natural Deduction for Intuitionistic Logic

Example

A proof that $\{A \vee B \text{ true}, A \supset C \text{ true}, B \supset C \text{ true}\} \vdash C \text{ true}$.

$$\dfrac{A \vee B \text{ true} \qquad \dfrac{A \supset C \text{ true} \qquad \dfrac{}{A \text{ true}}\,u}{C \text{ true}}\,\supset\!\mathrm{E} \qquad \dfrac{B \supset C \text{ true} \qquad \dfrac{}{B \text{ true}}\,w}{C \text{ true}}\,\supset\!\mathrm{E}}{C \text{ true}}\,\vee\mathrm{E}^{u,w}$$

Natural Deduction for Intuitionistic Logic

Inference rules for negation

Since we define negation by $\neg A := A \supset \bot$, for it we talk about **derived** inference.

Natural Deduction for Intuitionistic Logic

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- Formation rule

$$\frac{A\,\mathrm{prop}}{\neg A\,\mathrm{prop}}$$

Natural Deduction for Intuitionistic Logic

Inference rules for negation

Since we define negation by $\neg A := A \supset \bot$, for it we talk about **derived** inference.

- Formation rule

$$\frac{A\,\mathrm{prop}}{\neg A\,\mathrm{prop}}$$

- Introduction rule

$$\begin{array}{c}\overline{A\,\mathrm{true}}^{\,u}\\ \vdots \\ \dfrac{\bot\,\mathrm{true}}{\neg A\,\mathrm{true}}\,\neg\mathrm{I}^u\end{array}$$

Natural Deduction for Intuitionistic Logic

Inference rules for negation

Since we define negation by $\neg A := A \supset \bot$, for it we talk about **derived** inference.

- Formation rule

$$\frac{A\ \mathrm{prop}}{\neg A\ \mathrm{prop}}$$

- Introduction rule

$$\dfrac{\dfrac{}{A\ \mathrm{true}}\,u \\ \vdots \\ \bot\ \mathrm{true}}{\neg A\ \mathrm{true}}\ \neg\mathrm{I}^u$$

- Elimination rule

$$\frac{\neg A\ \mathrm{true} \qquad A\ \mathrm{true}}{\bot\ \mathrm{true}}\ \neg\mathrm{E}$$

Natural Deduction for Intuitionistic Logic

Examples

See many more examples in (Carr 2018).

Natural Deduction for Classical Logic

There are different ways to define a system of natural deduction for classic logic from the previous system of natural deduction for intuitionistic logic. For example, we can add the following inference rule for proof by contradiction (*reductio ad absurdum*) (see, e.g. (van Dalen [1980] 2013)).

$$\dfrac{\dfrac{}{\neg A\ \mathrm{true}}\,u \qquad \vdots}{} \qquad \dfrac{\bot\ \mathrm{true}}{A\ \mathrm{true}}\ \mathrm{RAA}^u$$

Natural Deduction for Classical Logic

Example

A proof that $A \vee \neg A \; \mathrm{true}$.

$$
\dfrac{\dfrac{}{\neg(A \vee \neg A)\;\mathrm{true}}u \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\neg(A \vee \neg A)\;\mathrm{true}}u \qquad \dfrac{\dfrac{}{A\;\mathrm{true}}w}{A \vee \neg A\;\mathrm{true}}\vee\mathrm{I}_2}{\bot\;\mathrm{true}}\neg\mathrm{E}}{\neg A\;\mathrm{true}}\neg\mathrm{I}^w}{A \vee \neg A\;\mathrm{true}}\vee\mathrm{I}_1}{\dfrac{\bot\;\mathrm{true}}{A \vee \neg A\;\mathrm{true}}\mathrm{RAA}^u}\neg\mathrm{E}
$$

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